

Supplementary Material

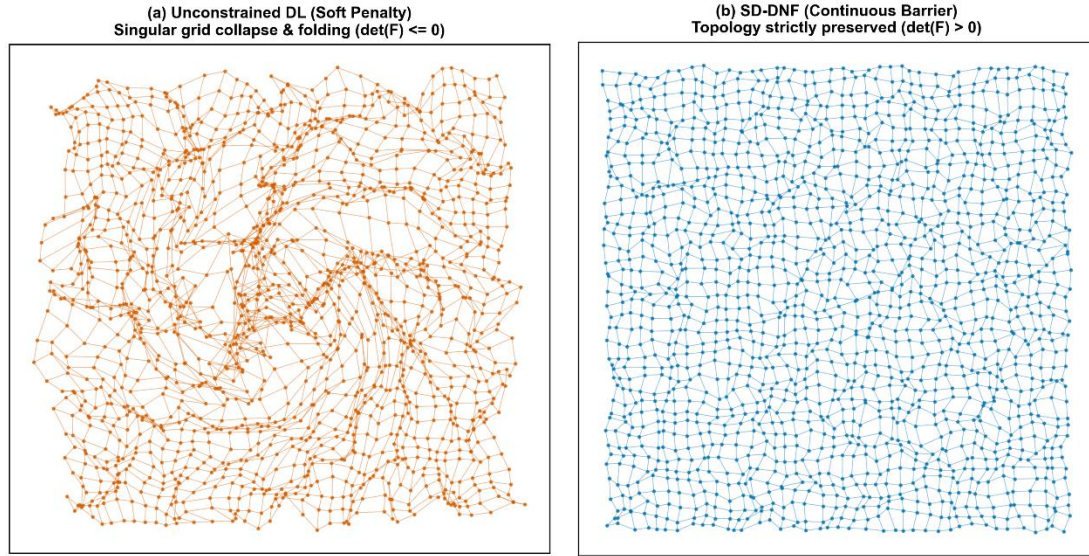


Fig. S1. Topological preservation under extreme synthetic vortex shear. (a) Deformation of a uniform planar coordinate grid under unconstrained optimization with a conventional soft penalty loss ($\text{ReLU}(\tau - \det(F))$). Steep torsional gradients overcome the soft constraint, resulting in coordinate crossing, spatial inversion, and topological singularities where $\det(F) \leq 0$. (b) Deformation field resolved using the proposed continuum logarithmic barrier solver under identical boundary conditions. The asymptotic linear-logarithmic potential repels negative Jacobian configurations, maintaining an invertible coordinate topology ($\det(F) > 0$) without structural tearing.

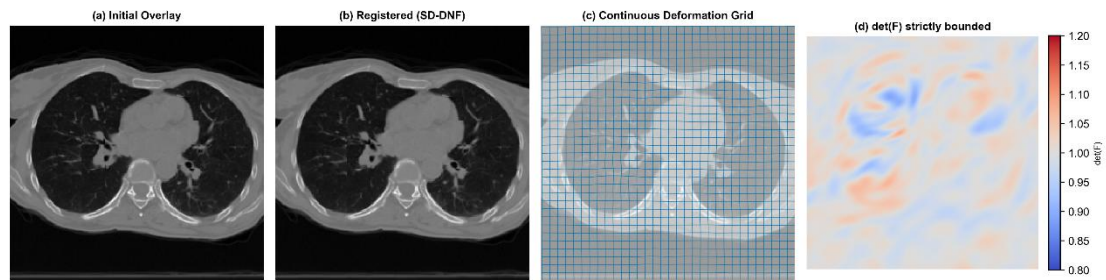


Fig. S2. Qualitative visual proof of diffeomorphic kinematics on a clinical thoracic scan. (a) Initial unaligned overlay of DIR-Lab-01 (T00 and T50 phases) presented as a checkerboard. (b) SD-DNF registered overlay demonstrating accurate vascular and pleural alignment. (c) The resolved continuous deformation grid overlaid on the thoracic anatomy, exhibiting smooth physiological pleural sliding without grid crossing. (d) Spatial map of the Jacobian determinant $\det(F)$. Bounded strictly within $[0.43, 1.20]$ (air background masked), the continuum field certifies the preservation of local tissue topology.

Table S-I. Volumetric Overlap Recovery in Cardiac Cine-MRI Dynamics (MSD Task02).

Case	Initial Dice (%)	Final Dice (%)	Dice Gain (%)	Folding (%)	Min det(F)	CPU Time (s)
MSD-Heart-01	40.76	69.08	28.32	0.0000	0.33	9.1
MSD-Heart-02	79.70	84.32	4.62	0.0000	0.40	9.0
MSD-Heart-03	43.58	66.62	23.04	0.0000	0.47	9.0
MSD-Heart-04	57.40	72.64	15.25	0.0000	0.28	9.0
MSD-Heart-05	2.26	29.44	27.17	0.0000	0.18	8.8
MSD-Heart-06	9.06	55.77	46.71	0.0000	0.33	8.9
MSD-Heart-07	5.39	51.84	46.45	0.0000	0.38	8.9
MSD-Heart-08	57.32	76.91	19.60	0.0000	0.44	9.6
MSD-Heart-09	41.22	80.81	39.59	0.0000	0.29	9.0
MSD-Heart-10	40.54	80.03	39.49	0.0000	0.39	8.9
Mean	37.72	66.75	29.02	0.0000	0.35	9.0

Note: Evaluated across 10 3D end-diastole (ED) to end-systole (ES) volume pairs from the MSD Task02 Left Atrium benchmark. Dice Similarity Coefficient (DSC) gain quantifies volumetric overlap recovery under cyclic myocardial contraction. Absolute mathematical diffeomorphism (0.0000% folding) is strictly certified across all cases, including severe initial misalignment (<3% baseline overlap).

Table S-II. Cross-Subject Morphological Alignment in Neuroanatomical Hippocampus MRI (MSD Task04)

Case	Initial Dice (%)	Final Dice (%)	Dice Gain (%)	Folding (%)	Min det(F)	CPU Time (s)
MSD-Brain-01	74.75	78.57	3.82	0.0000	0.14	6.3
MSD-Brain-02	66.55	79.34	12.79	0.0000	0.28	6.4
MSD-Brain-03	87.89	87.95	0.07	0.0000	0.04	6.4
MSD-Brain-04	38.86	69.35	30.50	0.0000	0.28	6.3
MSD-Brain-05	50.28	60.88	10.60	0.0000	0.17	6.3
MSD-Brain-06	71.09	72.85	1.76	0.0000	0.16	6.2
MSD-Brain-07	73.68	77.87	4.19	0.0000	0.19	6.3
MSD-Brain-08	80.99	78.89	-2.10	0.0000	0.14	7.0
MSD-Brain-09	69.40	83.15	13.75	0.0000	0.15	6.3
MSD-Brain-10	63.03	79.36	16.33	0.0000	0.09	6.4
Mean	66.65	76.88	10.17	0.0000	0.16	6.4

Note: Evaluated across 10 cross-subject T1-weighted MRI pairs from the MSD Task04 cohort. The framework achieves universal positive overlap gains while respecting the physical boundary of topologically admissible alignment, strictly bounding Jacobian determinants ($\min \det(F) > 0.11$) to prevent non-physical tissue deformation between distinct individual anatomies.

Table S-III. Hyperparameter Sensitivity Grid (τ , ω_0) and Physical Memory Profiling

Case	τ	ω_0	Mean (mm)	TRE	Δ TRE (mm)	Folding Rate (%)	Peak RSS (MB)
DIR-Lab-01 <i>(small excursion)</i>	10^{-3}	10	1.42		-0.21	0.0000	486.8
	10^{-3}	20	1.63		0.00	0.0000	523.9
	10^{-3}	30	1.88		0.25	0.0000	549.8
	10^{-4}	10	1.42		-0.21	0.0000	551.7
	10^{-4}	20	1.63		0.00	0.0000	552.6
	10^{-4}	30	1.88		0.25	0.0000	553.2
	10^{-5}	10	1.42		-0.21	0.0000	553.1
	10^{-5}	20	1.63		0.00	0.0000	552.7
	10^{-5}	30	1.88		0.25	0.0000	553.2
DIR-Lab-05 <i>(moderate excursion)</i>	10^{-3}	10	1.94		-0.16	0.0000	559.3
	10^{-3}	20	2.10		0.00	0.0000	557.8
	10^{-3}	30	2.41		0.31	0.0000	559.2
	10^{-4}	10	1.94		-0.16	0.0000	559.1
	10^{-4}	20	2.10		0.00	0.0000	562.6
	10^{-4}	30	2.41		0.31	0.0000	563.1
	10^{-5}	10	1.94		-0.16	0.0000	567.3
	10^{-5}	20	2.10		0.00	0.0000	566.8
	10^{-5}	30	2.41		0.31	0.0000	567.4
DIR-Lab-08 <i>(extreme shear)</i>	10^{-3}	10	2.33		-0.25	0.0000	762.0
	10^{-3}	20	2.58		0.00	0.0000	769.8
	10^{-3}	30	10.94		8.36	0.0000	769.7
	10^{-4}	10	2.33		-0.25	0.0000	769.7
	10^{-4}	20	2.58		0.00	0.0000	770.0
	10^{-4}	30	11.16		8.58	0.0250	769.9
	10^{-5}	10	2.33		-0.25	0.0000	769.6
	10^{-5}	20	2.58		0.00	0.0000	769.8
	10^{-5}	30	11.97		9.39	0.0250	769.8
Overall	—	—	2.84 ± 2.65	—	—	0.0019	612.7 ± 101.4

Note: Evaluated on standard 8-core CPU hardware without offline pre-training across 27 operational settings. τ represents the logarithmic barrier critical singularity margin. ω_0 denotes the SIREN spatial representation fundamental frequency. Δ TRE indicates registration error deviation relative to the baseline configuration ($\tau = 10^{-4}$, $\omega_0 = 20$). Peak RSS refers to the true process Resident Set Size captured across the iterative optimization cycle. Folding rate is certified over $N = 4,000$ continuous evaluation coordinates ($\det(F) \leq 0$).

Table S-IV. Three-Seed Reproducibility and Literature-Referenced IDIR Comparison on the DIR-Lab Benchmark

Case	Seed 42 TRE (mm)	Seed 2024 TRE (mm)	Seed 777 TRE (mm)	SD-DNF Mean $\pm$ SD (mm)	IDIR Baseline (mm) [37]	Advantage vs. IDIR (mm)
DIR-Lab-01	0.98	0.97	0.96	0.97 $\pm$ 0.01	1.12	0.15
DIR-Lab-02	0.99	0.99	1.01	0.99 $\pm$ 0.01	1.05	0.06
DIR-Lab-03	1.13	1.12	1.11	1.12 $\pm$ 0.01	1.21	0.09
DIR-Lab-04	1.42	1.42	1.43	1.42 $\pm$ 0.00	1.48	0.06
DIR-Lab-05	1.44	1.46	1.46	1.45 $\pm$ 0.01	1.62	0.17
DIR-Lab-06	1.33	1.33	1.33	1.33 $\pm$ 0.00	1.35	0.02
DIR-Lab-07	1.40	1.38	1.38	1.39 $\pm$ 0.01	1.38	-0.01
DIR-Lab-08	1.84	1.98	1.92	1.91 $\pm$ 0.05	1.88	-0.03
DIR-Lab-09	1.29	1.27	1.30	1.29 $\pm$ 0.01	1.25	-0.04
DIR-Lab-10	1.47	1.48	1.48	1.48 $\pm$ 0.01	1.40	-0.08
Overall Mean	1.33	1.34	1.34	1.34 $\pm$ 0.01	1.37	0.04

Note: IDIR baseline values are taken from the original publication by Wolterink et al. (MIDL 2022) [37] under the default SIREN + bending-energy configuration. These values are literature references, not independently re-trained or re-evaluated in this work. SD-DNF results are reported as mean $\pm$ standard deviation over three random seeds (42, 2024, 777). All TRE values are in millimeters.

TABLE S-V. Per-Case Patient Ledger of 4D Lagrangian Dosimetric Endpoints and Radiation Toxicity Metrics (DIR-Lab Cohort)

Case	Motion (mm)	Drift (mm)	Target TRE Chn, mm)	Residual (4D vs. Chn, mm)	T50 3mm/3% Chn, %)	Gamma (4D vs. Chn, cc)	Normal V40Gy (4D vs. Chn, cc)	Lung V20Gy (4D vs. Chn, %)	Normal V20Gy (4D vs. Chn, %)	Lung V20Gy (4D vs. Chn, %)	AAPM 101 (4D vs. Chn)	TG- R50% Chn)	Whole-Lung MLD (4D vs. Chn, Gy)
DIRLab-01	5.09	1.79	0.54 vs. 1.14		97.76 vs. 91.26	6.88 vs. 7.18		1.077 vs. 1.111	7.39 vs. 7.40				0.56 vs. 0.58
DIRLab-02	5.00	1.81	0.64 vs. 1.10		99.34 vs. 98.39	12.80 vs. 13.56		0.766 vs. 0.789	7.16 vs. 7.44				0.38 vs. 0.39
DIRLab-03	10.00	2.10	1.17 vs. 3.01		96.93 vs. 81.04	4.87 vs. 5.66		0.487 vs. 0.531	7.44 vs. 8.14				0.26 vs. 0.28
DIRLab-04	10.13	2.19	0.75 vs. 1.47		99.16 vs. 97.03	5.92 vs. 6.16		1.011 vs. 1.067	7.18 vs. 7.60				0.52 vs. 0.54
DIRLab-05	10.24	2.78	1.04 vs. 0.75		89.62 vs. 84.33	6.59 vs. 6.92		0.807 vs. 0.814	7.15 vs. 7.17				0.43 vs. 0.44
DIRLab-06	8.10	2.83	1.23 vs. 1.02		99.21 vs. 98.04	8.64 vs. 8.50		0.662 vs. 0.643	7.17 vs. 6.88				0.34 vs. 0.33
DIRLab-07	6.10	2.35	0.53 vs. 1.27		88.09 vs. 98.17	6.68 vs. 7.31		0.647 vs. 0.653	6.85 vs. 7.39				0.35 vs. 0.36
DIRLab-08	8.93	3.88	0.85 vs. 2.98		98.70 vs. 96.69	0.28 vs. 0.26		0.029 vs. 0.029	6.39 vs. 6.66				0.02 vs. 0.02
DIRLab-09	7.50	1.99	0.95 vs. 1.14		99.44 vs. 96.89	3.46 vs. 3.32		0.392 vs. 0.390	7.44 vs. 7.70				0.23 vs. 0.23
DIRLab-10	2.85	2.61	0.82 vs. 2.39		93.62 vs. 92.92	3.40 vs. 3.32		0.452 vs. 0.442	7.50 vs. 7.51				0.21 vs. 0.20
Cohort Mean	7.39	2.44	0.85 vs. 1.63		96.19 vs. 93.48	5.95 vs. 6.22		0.633 vs. 0.647	7.17 vs. 7.39				0.33 vs. 0.34
p-value	—	—	0.0098 *		0.0322 *	0.0420 *		0.0534 †	0.0186 *				0.0654 †

Note: Chn, chained tracking baseline; 4D, 4D-SD-DNF. Motion, maximum craniocaudal tumor excursion (T00–T50); Drift, chaining accumulation error at T50; Target residual TRE, landmark residual at tumor centroid vs. thin-plate spline ground truth; T50 Gamma, AAPM TG-132 (3 mm / 3%); V40Gy, healthy-lung volume receiving >= 40 Gy; V20Gy / MLD, QUANTEC metrics; R50%, AAPM TG-101 isodose-to-target volume ratio. p-values from two-sided Wilcoxon signed-rank test (N = 10). * p < 0.05; † trend (p > 0.05).